

AI-Driven-Crypto-Cybercrime-Intelligence

An AI-driven forensic intelligence framework for cryptocurrency transaction analysis that integrates Machine Learning, Graph Convolutional Networks (GCN), and Behavioral Analytics to support cybercrime intelligence generation and digital forensic investigations.

Overview

Cryptocurrencies have become increasingly associated with illicit activities such as money laundering, ransomware payments, darknet market transactions, and financial fraud. Traditional investigation approaches often struggle to process the scale and complexity of blockchain transaction networks.

This project proposes an end-to-end forensic intelligence framework capable of:

- Detecting illicit cryptocurrency transactions.
- Classifying cybercrime typologies.
- Uncovering hidden relationships within transaction networks.
- Prioritizing suspicious transactions for investigation.
- Generating actionable intelligence to support investigators and law enforcement agencies.

Framework Architecture

The proposed framework consists of four major components:

1. Supervised Classification

Random Forest was used as the benchmark binary classifier to distinguish between licit and illicit transactions.

Alternative baseline models, including K-Nearest Neighbours (KNN), were evaluated during experimentation.

2. Graph Intelligence (GCN)

Graph Convolutional Networks were employed to exploit blockchain transaction topology and propagate intelligence to previously unlabeled transactions.

3. Behavioral Analysis

Behavioral indicators derived from cybercrime literature were used to identify transaction typologies, including:

- Ransomware
- Money Laundering
- General Illicit Activity

4. Composite Risk Scoring

A weighted forensic risk score was developed to prioritize high-risk transactions:

$$\text{CRS} = 0.4 \times \text{RF} + 0.4 \times \text{GCN} + 0.2 \times \text{Behavioral Analysis}$$

Datasets

This study utilized publicly available blockchain datasets:

- Elliptic Bitcoin Dataset
- BitcoinHeist Ransomware Dataset
- AMLSim Synthetic AML Dataset

Experimental Results

Random Forest (Binary Classification)

- Accuracy: 98.7%
- Precision (Illicit): 99%
- Recall (Illicit): 87%
- F1-Score (Illicit): 93%
- ROC-AUC: 99.2%

KNN Baseline

- Accuracy: 97.57%
- Precision: 87.9%
- Recall: 87.13%
- F1-Score: 87.51%
- ROC-AUC: 97.64%

Graph Convolutional Network (GCN)

- Accuracy: 95.1%
- Precision: 94.8%
- Recall: 95.1%
- F1-Score: 94.9%

The proposed framework successfully extended intelligence to more than 157,000 previously unlabeled blockchain transactions.

Forensic Outputs

The framework generates investigation-ready outputs, including:

- Transaction ID
- Composite Risk Score (CRS)
- Suspicious / Non-Suspicious Classification
- Likely Cybercrime Typology
- Behavioral Justification
- Graph-Based Intelligence
- Priority Investigation Ranking

Research Contributions

- Integrated machine learning, graph intelligence, and behavioral analytics into a unified forensic framework.
- Demonstrated improved cryptocurrency crime detection performance.
- Enabled intelligence propagation to unlabeled blockchain entities.
- Produced actionable investigation leads suitable for cybercrime intelligence environments.

Future Work

Potential future enhancements include:

- Multi-blockchain support (Ethereum, Monero, DeFi).
- Temporal and Dynamic GCN architectures.
- Adversarial robustness evaluation.
- Explainable AI integration.
- Real-time intelligence deployment.

Citation

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How to Run

For detailed Kaggle API setup instructions, please refer to:

Kaggle_API_Setup_Guide.pdf